Case study

Code:

```
# Hidden Markov Model for user engagement
from hmmlearn import hmm
import numpy as np

# 3 hidden states: Interested, Neutral, Disengaged
model = hmm.CategoricalHMM(n_components=3, n_iter=100)

# Equal starting probability for all states
model.startprob_ = np.array([0.33, 0.33, 0.34])

# More balanced transition matrix
model.transmat_ = np.array([
    [0.5, 0.3, 0.2],  # Interested  → can transition to Neutral/Disengaged
    [0.2, 0.5, 0.3],  # Neutral     → can transition either way
    [0.1, 0.3, 0.6]   # Disengaged  → tends to stay disengaged
])

# Clearer distinction between states
model.emissionprob_ = np.array([
    [0.9, 0.1],  # Interested:  very likely to click
    [0.5, 0.5],  # Neutral:     equal chance
    [0.1, 0.9]   # Disengaged:  very unlikely to click
])

# Observed sequence: 0=click, 1=no-click
observations = np.array([[0, 1, 1, 0]]).T

hidden_states = model.predict(observations)
```

```
# Map state numbers to labels

state_labels = {0: "Interested", 1: "Neutral", 2: "Disengaged"}

labeled_states = [state_labels[s] for s in hidden_states]


print("Predicted Hidden States (raw)   :", hidden_states)

print("Predicted Hidden States (labeled):", labeled_states)
```

OUTPUT:

```
===== RESTART: C:\Users\CBIT\OneDrive\Documents\Desktop\099casestudyacc.py =====
Predicted Hidden States (raw)   : [0 2 2 1]
Predicted Hidden States (labeled): ['Interested', 'Disengaged', 'Disengaged', 'N
eutral']
```